

Second Partial

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Simplifications using algebra

$$f(x) = \frac{x}{(x^2 + 1)^{3/2}}$$

$$f'(x) = \frac{1}{(x^2 + 1)^3} \left[(x^2 + 1)^{3/2} - 3x^2 \sqrt{x^2 + 1} \right]$$

$$f'(x) = \frac{1 - 2x^2}{\sqrt{(x^2 + 1)^5}}$$

Hint: Use a change of variable to simplify the expression to the eye.

Change of variable

Since $x^2 + 1$ repeats, let's introduce the following change of variable $a = x^2 + 1$.

$$f'(x) = \frac{1}{(x^2 + 1)^3} \left[(x^2 + 1)^{3/2} - 3x^2 \sqrt{x^2 + 1} \right] \rightarrow f'(x) = \frac{1}{a^3} \left[a^{3/2} - 3x^2 \sqrt{a} \right]$$

Use properties of exponents

Recalling the following properties of exponents (and notation): $\sqrt{x} = x^{1/2}$, $x^{n \cdot m} = (x^n)^m$ we can do the following,

$$\begin{aligned} f'(x) &= \frac{1}{a^3} \left[a^{3/2} - 3x^2 \sqrt{a} \right] \\ &= \frac{1}{a^3} \left[\left(a^{1/2} \right)^3 - 3x^2 a^{1/2} \right] \end{aligned}$$

Factor and use properties of exponents

Factor $a^{1/2}$.

$$\begin{aligned} f'(x) &= \frac{1}{a^3} \left[\left(a^{1/2} \right)^3 - 3x^2 a^{1/2} \right] \\ &= \frac{a^{1/2}}{a^3} \left[\left(a^{1/2} \right)^2 - 3x^2 \right] \end{aligned}$$

Use properties of exponents: $x^n = 1/x^{n-1}$, $x^n x^m = x^{n+m}$ and $(x^n)^m = x^{n \cdot m}$.

$$\begin{aligned} f'(x) &= \frac{1}{a^3 a^{-1/2}} \left[\left(a^{1/2} \right)^2 - 3x^2 \right] \\ &= \frac{1}{a^{3-1/2}} \left[a^{2/2} - 3x^2 \right] \end{aligned}$$

Finish the arithmetic and change notation

$$f'(x) = \frac{1}{a^{3-1/2}} \left[a^{2/2} - 3x^2 \right]$$

Since $3 - \frac{1}{2} = \frac{5}{2}$ and $\frac{2}{2} = 1$,

$$\begin{aligned} f'(x) &= \frac{1}{a^{5/2}} [a - 3x^2] \\ &= \frac{1}{\sqrt{a^5}} [a - 3x^2] \end{aligned}$$

Undo the change of variable

Now we can undo the change of variable, since we can't do anything else, $a = x^2 + 1$.

$$f'(x) = \frac{1}{\sqrt{a^5}} [a - 3x^2] \rightarrow f'(x) = \frac{1}{\sqrt{(x^2 + 1)^5}} [x^2 + 1 - 3x^2]$$

Final step: Simplify

$$\begin{aligned} f'(x) &= \frac{1}{\sqrt{(x^2 + 1)^5}} [1 - 2x^2] \\ &= \frac{1 - 2x^2}{\sqrt{(x^2 + 1)^5}} \end{aligned}$$

$$f'(x) = \frac{1 - 2x^2}{\sqrt{(x^2 + 1)^5}}$$

Simplifications using algebra

Now, let's see how to reduce the following expression,

$$g(x) = 3 \left(\frac{1}{2} \right)^{x/2}$$

$$g'(x) = -3 \left(\frac{1}{2} \right)^{x/2} \ln \left[2^{1/2} \right]$$

$$g'(x) = -2^{-1-x/2} \ln [8]$$

Using properties of exponents

We start with the base of the exponential function, $\left(\frac{1}{2}\right)^{x/2}$

$$\begin{aligned}g'(x) &= -3 \left(\frac{1}{2}\right)^{x/2} \ln \left[2^{1/2}\right] \\&= -3 \frac{1^{x/2}}{2^{x/2}} \ln \left[2^{1/2}\right] \\&= -3 \frac{1}{2^{x/2}} \ln \left[2^{1/2}\right] \\&= -3 2^{-x/2} \ln \left[2^{1/2}\right]\end{aligned}$$

Using properties of logarithms

Let's move on with the logarithm, $a \ln [x] = \ln [x^a]$,

$$\begin{aligned} g'(x) &= -32^{-x/2} \ln [2^{1/2}] \\ &= -2^{-x/2} \ln \left[\left(2^{1/2} \right)^3 \right] \end{aligned}$$

Using properties of exponents

We are going to do an exponent exchange using the properties of exponents ($(a^n)^m = a^{n \cdot m} = (a^m)^n$). Recalling the expression, $g'(x) = -2^{-x/2} \ln \left[(2^{1/2})^3 \right]$ We are going to focus on $(2^{1/2})^3$.

$$(2^{1/2})^3 = 2^{3/2} = (2^3)^{1/2} = 8^{1/2}$$

Replace

Now that we know $(2^{1/2})^3 = 8^{1/2}$

$$\begin{aligned} g'(x) &= -2^{-x/2} \ln \left[(2^{1/2})^3 \right] \\ &= -2^{-x/2} \ln \left[8^{1/2} \right] \end{aligned}$$

The last step is to use properties of logarithms and exponents to get the expression $g'(x) = -2^{-1-x/2} \ln [8]$.

Properties of logarithms

Using the same property as before, $a \ln [x] = \ln [x^a]$, we can exchange the exponent of argument of the logarithm as a factor

$$\begin{aligned} g'(x) &= -2^{-x/2} \ln [8^{1/2}] \\ &= -2^{-x/2} \frac{1}{2} \ln [8] \end{aligned}$$

Properties of exponents

Using the following properties, $x^{-n} = 1/x^n$, $x = x^1$ and $x^n \cdot x^m = x^{n+m}$

$$\begin{aligned} g'(x) &= -2^{-x/2} \frac{1}{2} \ln[8] \\ &= -\frac{1}{2^{x/2}} \frac{1}{2} \ln[8] \\ &= -\frac{1}{2^{x/2+1}} \ln[8] \\ &= -2^{-x/2-1} \ln[8] \end{aligned}$$

End

$$f'(x) = \frac{(x^2 + 1)^{3/2} - 3x^2 \sqrt{x^2 + 1}}{(x^2 + 1)^3}$$

$$f'(x) = \frac{1 - 2x^2}{\sqrt{(x^2 + 1)^5}}$$

$$g'(x) = -3 \left(\frac{1}{2} \right)^{x/2} \ln \left[2^{1/2} \right]$$

$$g'(x) = -2^{-x/2-1} \ln [8]$$